

Vehicle dynamics modelling for the National Advanced Driving Simulator

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Abstract: This paper provides an overview of the development of a vehicle dynamics simulation model for use on the National Highway Traffic Safety Administration's (NHTSA's) National Advanced Driving Simulator. The paper describes fundamental aspects of models used to represent rigid body chassis and suspension systems, powertrain, tyres, brakes, steering and aerodynamics. Representative data from laboratory measurements, instrumented field tests and simulation runs of a 1997 Jeep Cherokee Sport are presented to illustrate simulation development and validation efforts. A companion paper to this one also uses Jeep Cherokee data. Both papers highlight current capabilities and methodologies employed by two organizations that have worked, often collaboratively, to advance the state of the art of vehicle dynamics modelling and simulation validation. This paper features work done by NHTSA's Vehicle Research and Test Center; the companion paper reports on work done by Systems Technology, Incorporated (STI).

Keywords: National Advanced Driving Simulator, simulator, simulation, vehicle dynamics, model, validation

1 INTRODUCTION

The National Highway Traffic Safety Administration (NHTSA) has funded development of the National Advanced Driving Simulator (NADS), a driver-in-the-loop, high fidelity and real-time driving simulator. The NADS, shown in Fig. 1, is the most advanced driving simulator in the world and it has been designed to immerse the test subject (driver) fully into the driving scenario. The NADS provides a safe, efficient tool for driver impairment studies, for driver workload and adaptation studies, and it can be used to support development of automotive safety regulations [1].

Four vehicles have been modelled and are currently available for use on NADS: a Ford Taurus, a Chevrolet Malibu, a Jeep Cherokee and a tractor-semitrailer. One of the vehicles is mounted in the NADS dome. For the light vehicles, the entire vehicle is placed in the dome; for the heavy truck just the tractor cab plus front hood is used. The dome contains a visual system that uses 15 projectors to provide the test subject with a 360° field of

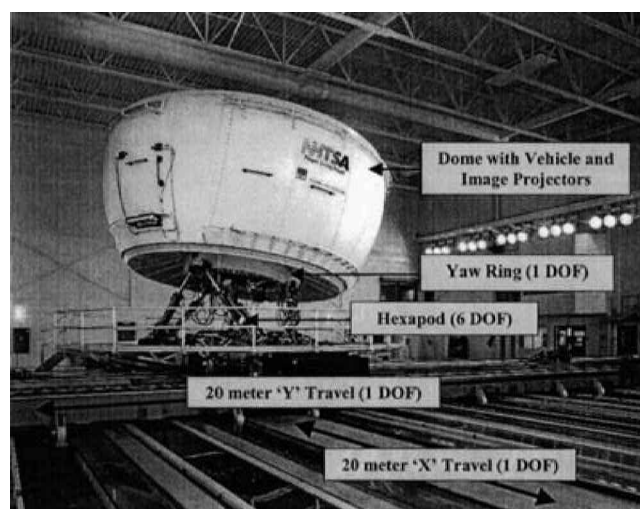


Fig. 1 National Advanced Driving Simulator

view. Current visual and terrain scenarios include urban, rural, residential, mountainous and other environments that can be populated with traffic, pedestrians and controlled traffic signals and signage. The NADS contains a high fidelity audio system that provides the test subject with realistic audio feedback from the vehicle and environment. Terrain (road) features such as surface

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type and friction, wet, icy, etc. and terrain profiles such as slopes, cross grades, potholes, curbs, etc. are modelled and synchronized with the visual and audio cues.

The vehicle dynamics simulation computes the response of the vehicle as a result of the test subject (driver) inputs for steering wheel motion, throttle displacement, brake pedal displacement, clutch engagement and transmission gear position. Each of the NADS vehicles is fitted with the NADS Control Feel System (CFS) hardware that provides driver feedback through the driver controls: steering wheel, throttle, brakes, clutch and gearshift mechanisms. The terrain features and profiles are modelled as inputs to the vehicle dynamics simulation. In addition, the vehicle dynamics simulation computes the aerodynamic forces arising from simulated steady and gusting winds and those caused by passing vehicles.

The responses computed by the vehicle dynamics simulation include the position, velocity and acceleration states of the vehicle. The visual system image generators that project the real-time images of the driving scenario use these variables. These variables are also used to determine the position of the vehicle on the terrain. They are also used to drive the NADS motion system, which moves the dome to replicate the motions of an actual vehicle driving through the simulated scenario.

The NADS motion system has nine degrees of freedom as indicated on Fig. 1. It has a 330° range of motion yaw ring between its six-degree-of-freedom hexapod and the dome containing the vehicle. The hexapod is mounted on a $20\text{ m} \times 20\text{ m}$ X - Y motion system that provides for large horizontal excursions and two additional degrees of freedom.

To simulate high frequency road disturbances and high frequency loads through the tyres and suspension, NADS contains four vibration actuators, mounted at points of suspension–chassis interaction. These vibration actuators are mounted between the floor of the dome and vehicle, and they act only in the bounce direction of the chassis. The vibration actuators were designed with vibration isolator counterweights to minimize vibrations transmitted to the dome. The vibration actuators can be driven on the basis of forces computed from the vehicle dynamics simulation and/or on the basis of random forces to mimic road disturbances.

Realistic mathematical vehicle dynamics models are a prerequisite for successful NADS operation. This paper highlights efforts made by NHTSA's Vehicle Research and Test Center (VRTC) to provide such models. In addition to describing general aspects of the NADS vehicle dynamics simulation, the paper focuses on required laboratory measurements to obtain meaningful parametric values for vehicle models, instrumented field testing for full model evaluation and simulation model validation. The overall methodology presented is a culmination of over a decade of effort by VRTC to refine and streamline simulation development and evaluation.

A series of papers have been written describing details of the light vehicle models [2–5] and their validation [6–8]. Papers describing the heavy truck model and its validation are forthcoming. This paper offers more of an overview rather than focusing on particular details of the models. The Jeep Cherokee model and its validation are used here as an exemplar of the overall processes that were used for developing all vehicle models.

2 NADS VEHICLE DYNAMICS

A general purpose, multibody, dynamics program, Real Time Recursive Dynamics (RTRD), developed by the University of Iowa's Center for Computer-Aided Design, is the core dynamics code of NADS vehicle dynamics simulation, NADSdyna [9]. NADSdyna includes vehicle subsystems that interact with the RTRD. Six vehicle submodels are included in NADSdyna: powertrain, cruise control, tyres, brakes, steering and aerodynamics. These submodules provide NADSdyna with a broad range of functional capabilities and allow most production vehicles to be simulated. In the NADS application, the inputs to NADSdyna come from the test subject (driver) inputs, from the modelled terrain features and profile and from the modelled wind. The outputs from NADSdyna at each simulation time step are used to drive the vehicle through the scenario and to direct the motion, visual and audio systems to respond accordingly.

The RTRD is used to model the rigid body behaviour of chassis, wheels and suspension and steering linkages. It offers great capabilities in terms of rigid body mechanism modelling and accounts for non-linear kinematic and dynamic effects, such as roll steer, roll centres and squat/lift forces. The RTRD is based on a topological analysis of a rigid body mechanism formulated using minimum relative joint coordinates. It can model either open or closed loop chains. Its constraint library supports revolute, spherical, translational, universal, spherical–translational, revolute–spherical and translational–revolute joints. For closed chains, distance constraints, spherical, and revolute–spherical cut-joint types are supported. Springs, dampers, antiroll bars and leaf springs are modelled as external forces. Non-linear elements can be used.

Two fundamental requirements of a vehicle dynamics simulation used with a driving simulator are that it executes faster than real time and that it provides sufficient fidelity to the actual vehicle to convince a simulator driver that they are driving the actual vehicle. These two requirements are conflicting in that higher simulation fidelity requires greater model complexity and longer run times. NADSdyna has been developed to meet both of these requirements. NADSdyna has been shown to provide simulated responses that capture the essence of vehicle behaviour necessary for realistic simu-

lator operation. The basic responses of the vehicle to driver inputs of steering, braking and throttle are adequately predicted, as are understeer characteristics. Tyre and suspension responses to varying terrains and road obstacles are also predicted by NADSdyna well enough to provide the driver with realistic feedback.

NADSdyna does not include models for bushing compliance. Instead, suspension compliance characteristics are added to suspension kinematics, as is typical of lumped parameter based simulation models. NADSdyna also does not include models for non-rigid bodies (e.g. the chassis is modelled as a rigid body). These modelling decisions were made so as not to lengthen simulation run times and not to make NADSdyna more complex than necessary for successful NADS operation.

The exemplar vehicle modelled and tested is a 1997 Jeep Cherokee Sport. It has a 4.0 l I6 engine, four-speed automatic transmission, cruise control, pitman arm power-assisted steering, four-wheel antilock brakes and P255/75R15 Goodyear Wrangler RT/S OWL tyres with cold inflation pressure of 36 lbf/in².

2.1 Multibody dynamics modelling

A thorough exposure to the multibody dynamics formalisms used by the RTRD goes beyond the scope of this document. Interested parties should refer to specialized references. The RTRD has a recursive formulation that requires rigid bodies to be connected in an open loop structure, or a spanning tree [10, 11]. The Jeep model contains six closed loops. A joint on each loop is cut to obtain a spanning tree and constraint equations associated with Lagrange multipliers are imposed at these cut joints to force kinematically admissible motion. The procedure cuts joints that will result in a model with optimum computational efficiency characterized by a minimum number of coordinates and constraint equations [12]. The overall RTRD model for the Jeep Cherokee is shown in Fig. 2. Table 1 summarizes the rigid bodies and joints modelled.

VRTC subcontracted with SEA, Incorporated, to measure suspension kinematics and compliances, shock absorber characteristics, vehicle and suspension component geometry and centre-of-gravity and inertial properties of the vehicle and suspension components [13]. SEA, Incorporated, used its Vehicle Inertia Measurement Facility to determine vehicle weight, centre-of-gravity longitudinal, lateral and height positions, vehicle roll mass of inertia, pitch mass of inertia, yaw mass of inertia and roll/yaw mass product of inertia. The suspension linkages' inertial properties were measured using bifilar and trifilar pendulums. The suspension kinematic and compliance measurements were made by SEA using in-house equipment. The shock absorbers were tested at Detroit Testing Labs.

The measurements were used to provide parameters

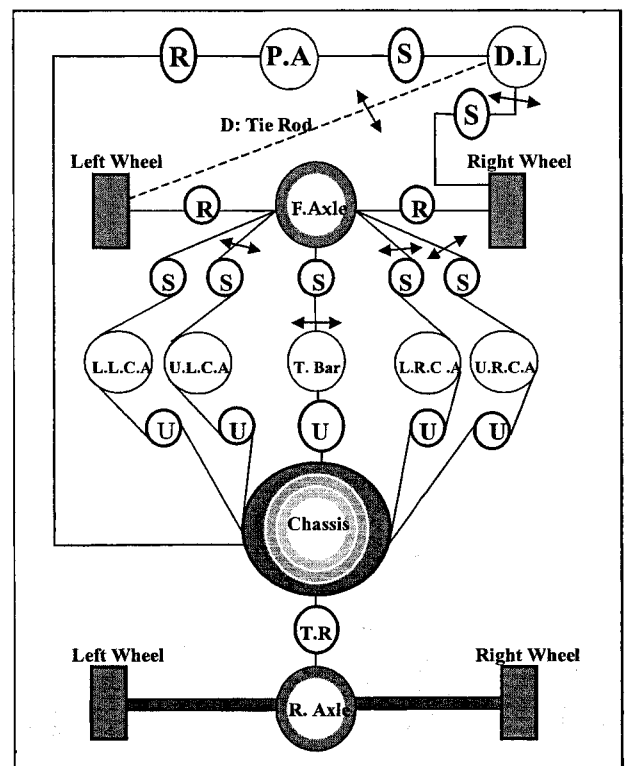


Fig. 2 RTRD multibody model of the Jeep Cherokee

Table 1 Rigid bodies and joints used to model the Jeep Cherokee

Rigid bodies	Comments
PA	Pitman arm
DL	Drag link
F axle	Front axle
LLCA	Lower left control arm
ULCA	Upper left control arm
LRCA	Lower right control arm
URCA	Upper right control arm
T bar	Track bar
R axle-left and right wheel	Rear axle and two attached wheels as one rigid body
Front left wheel	Front left wheel independent of the front axle
Front right wheel	Front right wheel independent of the front axle
Chassis	Chassis, passengers, drivers, and cargo as one rigid body

Joints	Comments	Number used
S	Spherical	7
R	Revolute	3
U	Universal	5
D	Distance constraint	1
TR	Translational revolute composite joint	1
↔	Cut constraint	6

used in the NADSdyna model/simulation. Also, the suspension kinematic tests were used to verify the accuracy of the multibody suspension model. The model was exercised in bounce and roll, and wheel steer and camber angle motions were checked against measured suspension characteristics. Simulation predictions of overall

front and rear roll stiffness and ride rates were also checked against measured responses. These types of checks are vital parts of evaluating the overall simulation. They provide confidence that the model is formulated correctly and that the parameters used are correct. These sanity checks can uncover problems that might otherwise go unnoticed or lead to problems later on when comparing against field test data. Experience has taught the present authors that it is better to identify model and/or parameter problems early in model development than when the full simulation, with all of its submodels and their complex interactions, is being used.

Other sanity checks were performed prior to attempting to validate against field test data. The simulated wheelbase and track widths were confirmed, as was Ackerman steering behaviour. The simulated wheel loads were checked against vehicle weight, and steady state lateral and longitudinal load transfers were checked against simple theory.

2.2 Suspension force elements

The Cherokee front suspension has a solid axle and coil springs. Coil spring and bump stop characteristics are consolidated and modelled using translational spring-damper-actuator (TSDA) elements. The top graph of Fig. 3 shows measured force-deflection curves that are used in non-linear TSDA models for front suspension springs and bump stops. TSDA elements use cubic spline interpolation to determine spring force on the basis of deflection. The line of action of the force is determined by spring end point positions.

The Cherokee rear suspension contains a solid axle and leaf springs. As shown in the bottom graph of Fig. 3, leaf springs exhibit greater hysteresis than coil springs. The leaf spring model used in NADSdyna is adopted from Fancher *et al.* [14] and accounts for hysteresis. Spring force is empirically formulated as a first-order differential equation:

$$\frac{\partial F}{\partial \delta} = \frac{F_{\text{env}} - F}{\beta} \quad (1)$$

where

- F spring force
- F_{env} upper or lower boundary envelope force
- δ spring deflection
- β decay constant (can differ for increasing/decreasing load)

For sufficiently small time steps, a valid solution is

$$F_i = F_{\text{env}i} + (F_{i-1} - F_{\text{env}i}) \exp(-|\delta_i - \delta_{i-1}|/\beta) \quad (2)$$

where subscript i indicates the current time step and $i-1$ the previous time step. The upper and lower envelopes are computed using cubic spline interpolation. This model is good for up to 25 Hz of road disturbance input

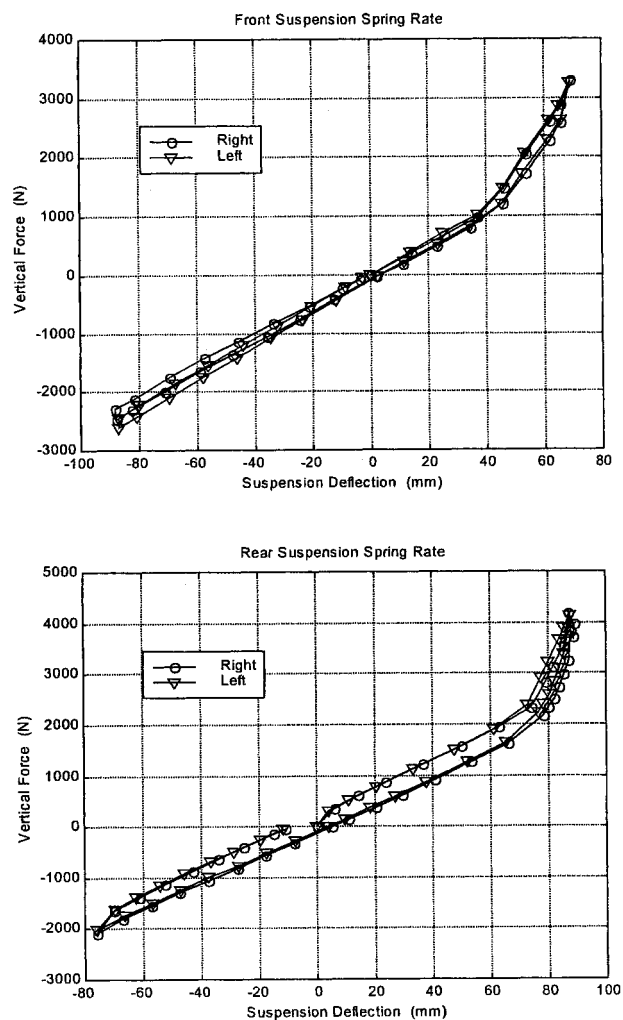


Fig. 3 Front and rear spring and bump stop characteristic curves

[14, 15]. In NADSdyna, leaf spring forces are applied at the line of action between two connecting points on the chassis and axle.

Shock absorbers in the front and rear suspensions and the steering shock absorber were modelled using TSDA cubic spline curve elements that relate damping forces to relative velocities between two rigid bodies connecting each shock. Figure 4 shows measured front shock absorber characteristic curves and the fit to data that was used in the NADSdyna model.

The damping is dependent on excitation frequency. As frequency increases, hysteresis loop sizes increase. The method used in NADSdyna does not model this effect; it is based on a nominal frequency (1–2 Hz). Modelling the damper extensively goes beyond the simulator's objective and would add unnecessary complexities to modelling efforts. The force-velocity relation used is sufficient for the needs of NADS.

The Cherokee's front and rear suspensions have anti-roll (stabilizer) bars. Antiroll bars have a large influence on lateral load transfer, understeer characteristics and

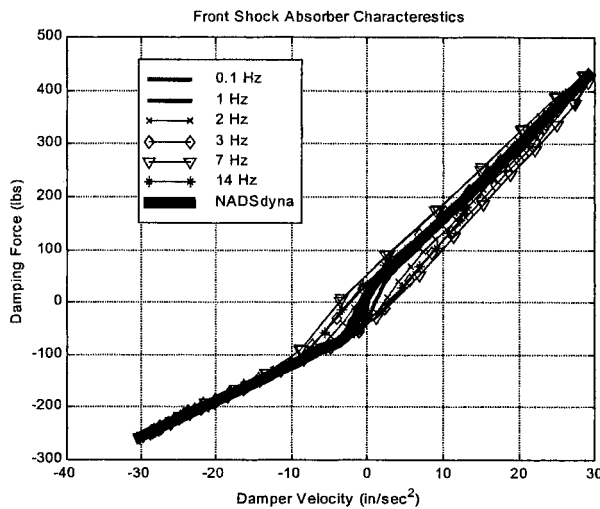


Fig. 4 Front shock absorber characteristic curves and curve fit used in NADSDyna modelling

vehicle handling and stability. In NADSDyna models, the physical geometry of the antiroll bars is not modelled with the multibody dynamics model. Rather, vertical forces are applied to the spindles as a function of roll angle to account for antiroll bar effects, as explained in reference [16]. This modelling method was selected to reduce the computational burden caused by including more bodies and stiff springs.

Roll stiffness of the antiroll bar is computed from overall stiffness by deducting roll stiffness due to suspension springs and tyre stiffnesses. The antiroll bar stiffness and suspension stiffness are in parallel and tyre stiffness is in series with their composite. The front and rear antiroll bar stiffness is computed as follows:

$$K_{aux} = \frac{K_{\phi} K_w (T_s^2/2) + K_{\phi} K_T (T_r^2/2) - K_w K_T T_s^2 (T_r^2/4)}{K_T (T_r^2/2) - K_{\phi}} \quad (3)$$

where

K_{ϕ}	measured overall suspension roll stiffness
K_w	measured suspension vertical stiffness
K_T	measured tyre vertical stiffness
T_s	suspension springs lateral distance
T_r	track width

2.3 Suspension system compliance

Compliances result from flexibility and deflection of suspension components, mainly bushings. These deflections cause, most importantly, additional steer and camber angles to be produced by road wheels which affect under-steer properties and consequently lateral/directional handling and stability. Static and dynamic stiffness characteristics of bushings are non-linear and much stiffer than other compliant elements used in a suspension. This compliance can be modelled either by chang-

ing steer or camber angles quasi-statically or by including stiff translational and rotational springs at bushing locations. Quasi-static implementation is used in NADSDyna since compliance effects have high frequency. Their inclusion would result in a rigid body dynamics that required a dramatically smaller integration time step, impeding real-time computing.

Front and rear aligning moment and lateral force compliance steer, and front and rear lateral force compliance camber, are included in NADSDyna models. Steer and camber compliances are modelled as linear functions of load (lateral force and aligning moment) and added to steer and camber angles determined by suspension kinematics [16].

2.4 Vehicle subsystems modelling

For a complete vehicle model, the multibody formalism is augmented by other subsystems: powertrain, cruise control, wheels, brakes, tyres, steering and aerodynamic forces. Each of the subsystem models is coupled to the RTRD code and computations are completed for each of them at each simulation time step. This section describes the models of each subsystem for the 1997 Jeep Cherokee Sport.

2.4.1 Powertrain

The powertrain subsystem is based on a one-dimensional torque formulation that includes torque and angular velocity transmitted through each model component [9]. Cherokee components include engine, torque converter, automatic transmission gearbox, four-two-wheel drive transfer case, differential and final drive.

The engine is modelled using a quasi-static engine map that relates throttle input and engine speed to engine torque. This model neglects torque pulsation due to combustion in individual cylinders and also neglects variant-inertia torque due to reciprocating motion of the pistons. In the actual vehicle, these high frequency torque variations are filtered by the torque converter. In NADSDyna, a first-order lag is applied to throttle input to account for fuel system and throttle lags. The lag time constant can be tuned to match temporally the simulated engine's response to the actual vehicle's response. The engine formulation is

$$J\dot{\omega}_e = T_e - T_{ac} - T_{tci} \quad (4)$$

where

J	equivalent engine inertia
$\dot{\omega}_e$	engine angular acceleration
T_e	indicated engine torque
T_{ac}	accessory engine torque
T_{tci}	torque converter input torque

Engine torque is obtained from engine map surface data using bilinear interpolation of throttle input and engine

speed:

$$T_e = \Phi(\omega_e, \phi_f) \quad (5)$$

where

$$\begin{aligned} \phi_f & \text{throttle input} \\ \omega_e & \text{engine speed} \\ \Phi & \text{engine surface data relation} \end{aligned}$$

Accessory torque is torque lost because of engine friction and pumping losses. It is estimated as a linear function of engine speed:

$$T_{ac} = a_0 + a_1 \omega_e \quad (6)$$

where a_0 and a_1 are constants.

Torque input to the torque converter is

$$T_{tci} = \frac{\omega_e^2}{K^2} \quad (7)$$

where K is the torque converter capacity factor.

The torque converter is assumed inertialess and to be always operating at capacity. Therefore, it can be adequately modelled by its steady-state performance characteristics. These assumptions are deemed justifiable since studies of torque converter dynamics have shown the quasi-steady-state model to be adequate for engine velocity disturbances with frequency content less than twice per revolution [9].

The converter output torque passed to the transmission is

$$T_{tco} = \gamma T_{tci} \quad (8)$$

where γ is the torque ratio. Torque ratio and capacity factor are functions of input to output speed ratio (Fig. 5). They are obtained from torque converter characteristic curves using cubic spline interpolation:

$$K = \Gamma\left(\frac{\omega_e}{\omega_{ti}}\right) \quad (9)$$

and

$$\gamma = \Psi\left(\frac{\omega_e}{\omega_{ti}}\right) \quad (10)$$

The Cherokee automatic transmission has six ranges and shift lever positions. Park, reverse and neutral are conventional and mechanically operated. The '1-2', '3' and 'D' (drive) ranges provide electronically controlled shifting. Position '1-2' provides first and second gears only, position '3' provides first, second and third gears and 'D' provides first through fourth gears.

The transmission input shaft is connected to the torque converter, and the output shaft is linked to the differential. Gear shifting operation is determined on the basis of current gear position, throttle input, output shaft speed and a shift logic map. The gear ratios are listed in Table 2 with an assumed efficiency of 95 per cent.

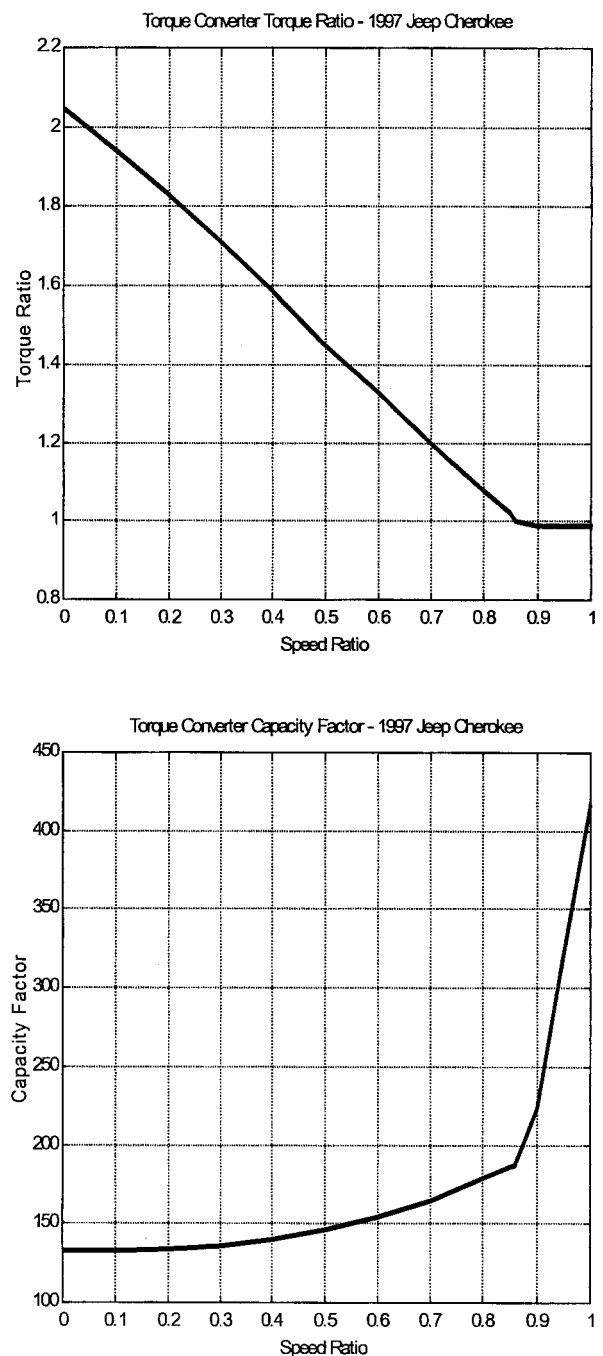


Fig. 5 Torque converter torque ratio and capacity factor

Table 2 Transmission gear ratios

Gear	Ratio
1st	2.80
2nd	1.55
3rd	1.00
4th	0.75
Reverse	-2.39

The shift logic map is shown in Fig. 6 where the output shaft speed is the speed of the shaft linked to the gear set.

The transmission is assumed inertialess, but blending functions at the time of shifting represent torque and speed transitions. Shapes for these functions are provided in reference [4]. These functions provide for smooth gear transitions in 1 s.

The Cherokee transfer case provides for part-time four-wheel drive. When engaged, both the front and the rear axles are mechanically locked together and turn at the same speed. The moments of inertia of the gear components are assumed negligible, torque distribution to the two output shafts is assumed symmetric and efficiency of the transfer case is approximated to be 95 per cent.

The transfer case input torque, T_T , is evenly proportioned between the two drive shafts:

$$T_{FS} = T_{RS} = \frac{1}{2} n_{tc} n_{ef} T_T \quad (11)$$

where

T_{FS}	front shaft torque
T_{RS}	rear shaft torque
n_{ef}	transfer case efficiency (0.95)
n_{tc}	transfer case ratio

The transmission provides for two four-wheel drive options: 4H with $n_{tc} = 1$ and 4L with $n_{tc} = 2.72$. For two-wheel drive, the drive train is rear-wheel drive with T_{RS} equal to T_T .

The differential gear set divides torque between right and left axle shafts. It allows axle shafts to turn at different speeds when turning corners. The Cherokee has a limited slip differential.

The NADSdyna differential model allows a variable torque split based on output shaft speed difference [9].

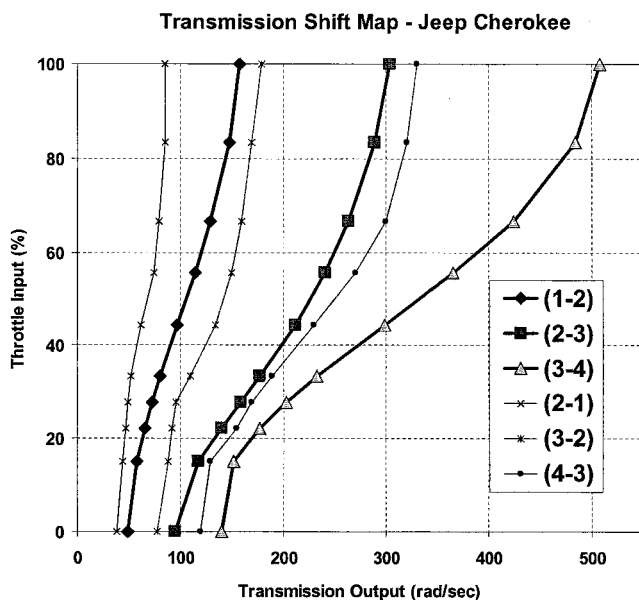


Fig. 6 Automatic shift logic map

This model can simulate a conventional differential by reducing bias torque to zero. The model assumes an inertialess differential, and that torque distribution is a function of speed difference only. Details of this model can be found in reference [4].

The ratio between the drive axle and the wheels is expressed by the following ratio and assumed efficiency: $N_{fd} = 3.55$ and $\eta_{fd} = 0.98$, where N_{fd} is the gear ratio of the final drive and η_{fd} is torque efficiency.

2.4.2 Cruise control

The vehicle speed control system is electronically controlled and vacuum operated. When engaged, this system takes over accelerator operations at speeds greater than 35 miles/h (60 km/h) and less than 85 miles/h (137 km/h). Controls are mounted on the steering wheel and consist of ON/OFF, SET, RESUME/ACCEL, CANCEL and COAST.

The cruise control model uses a proportional-integral controller to make errors between desired speed and actual vehicle speed tend towards zero for disturbances such as grade changes, aerodynamics, rolling resistance, etc. This model can be used with any conventional cruise control; only the driver interface changes. The basic controller mathematics is

$$e = u_d u \quad (12)$$

$$\phi_f = k_p e + k_i e/s \quad (13)$$

where

e	error between desired and actual speed
u_d	desired vehicle speed
u	actual vehicle speed
s	Laplace transform
ϕ_f	throttle command

These control equations are complemented with logical statements in NADSdyna so as to have the same cruise control commands as in the actual vehicle.

2.4.3 Wheel modelling

The road wheels are not modelled as rigid bodies; only the spin degree of freedom is formulated using

$$J\omega_w = T - F_x r \quad (14)$$

where

J	wheel spin inertia
T	applied torque
ω_w	wheel spin acceleration
F_x	tyre longitudinal force
r	rolling radius

The wheel orientation is the same as the suspension's spindle body.

The longitudinal and lateral slip equations are solved using first-order differential equations as suggested by Bernard and Clover [17] and presented in reference [4].

2.4.4 Tyre model

NADSdyna supports the System Technology, Incorporated (STI) tyre model [18], a look-up table tyre model and a user-defined interface tyre model. Using the user-defined interface subroutine, the user can implement any desired tyre model (e.g. Pajacka Magic Formula Model). The STI tyre model includes pertinent quasi-static tyre dynamics as well as lateral and longitudinal lag dynamics. Comparisons of tyre model predictions with test data are provided in the companion paper [19]. The tests performed on the Cherokee's P225/75R15 tyres to obtain data for the model included steering, cambering, braking/driving, combined and discrete sinusoidal (to measure tyre lag characteristics) [4].

2.4.5 Aerodynamic forces and moments

The SAE J1594 aerodynamic recommended practice is used to model aerodynamic forces [20]. The mathematical application is similar to reference [16]. NADSdyna also contains a model for wind gusts resulting from passing vehicles. The wind gust model is based on work by Heffley [21].

Aerodynamic force and moment coefficients for the 1997 Jeep Cherokee were measured at Lockheed Aeronautical Systems Company's Low Speed Wind Tunnel [22]. The vehicle was installed in the low speed test section on an external balance, four-point, support system. Data were acquired at nominal velocities of 45, 65 and 90 miles/h and at yaw angles spanning from 0° to 180° at 10° increments. The coefficients shown in Fig. 7 as a function of aerodynamic (wind) heading angle, β , include lift coefficient (CL), drag coefficient (CD), side

force coefficient (CS), pitch moment coefficient (CPM), yaw moment coefficient (CYM) and roll moment coefficient (CRM).

2.4.6 Brake system

Power-assisted front disc and rear drum brakes are standard equipment for the Cherokee. The 1997 Cherokee is equipped with a fixed rear pressure-proportioning valve. Brake system dynamics is modelled by including a first-order lag between brake pedal force and pressure at the brake lines. See reference [4] for the mathematical formulation used.

Measurements of brake pedal force, front line brake pressure and rear line brake pressure were done at VRTC. The vehicle was stationary, with engine running, as a cyclic increasing brake pedal force was slowly applied. The brake pedal force to front line brake pressure relation was obtained, and linear regression was used to determine curve data points used in NADSdyna. Rear brake pressure line to front brake pressure line proportioning was similarly determined. The knee point and slope (rear/front) were determined using linear regression at increasing pressures. Hysteresis was not considered for rear-to-front pressure proportioning data. The brake line pressure to brake torque relation is measured using VRTC's Road Transducer Plates. Brake torques were determined from measured tyre forces multiplied by wheel rolling radii. Linear regression was used to determine curve data points for front and rear brakes used in NADSdyna. The time delay between brake force and front brake line pressure is estimated from a brake force step input test to be 0.2 s.

The ABS model uses a proportional plus derivative controller to drive wheel slip control to a set point. Modulated pressure is computed individually for each wheel so that computations can proceed in parallel with those for wheel brake torque. Basic ABS controller formulations are explained in reference [3].

2.4.7 Steering system

On NADS, actual, measured steering wheel angles and rates are fed in real time as inputs to NADSdyna. NADSdyna computes vehicle responses and resulting forces acting on the steering 'rack'. For the Cherokee, 'rack' force is the force at the lower end of the pitman arm. Power steering effects are handled by NADS CFS software and hardware. CFS models steering feedback on the basis of steering forces and motions and includes a model for on-centre steering lash.

3 SIMULATION VALIDATION

The simulation validation methodology used for the NADSdyna Cherokee model has been developed over a number of years by VRTC [23]. The methodology con-

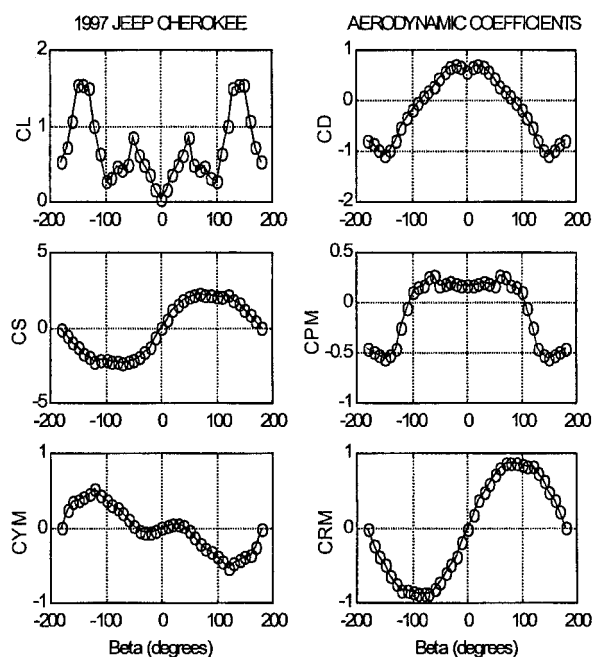


Fig. 7 Aerodynamic coefficients

sists of three main phases: experimental data collection from instrumented vehicle tests, vehicle parameter measurement and comparison of simulation predictions with experimental data. As mentioned, prior to comparing full simulation predictions with field test data, numerous sanity checks and independent checks of subsystem models are made. Trying to find sources of discrepancy between simulations and experiments by looking at the vehicle as one unit is often a hopeless effort owing to complex interactions between many subsystems.

Simulation validation is performed in both time and frequency domains. Experimental driver inputs (steering, braking and throttle) are used as inputs. Simulation predictions are compared with experimental outputs. Time domain validation is good for demonstrating that the simulation can correctly predict steady state conditions and that non-linear effects are properly modelled. High frequency transient phenomena, however, are difficult to study in the time domain. They are best examined through frequency domain studies.

3.1 Vehicle field testing

VRTC has done extensive vehicle testing to provide data for NADSdyna evaluation. The Cherokee instrumentation and field testing are similar to those used to test the Ford Taurus [24]. Vehicle field testing minimized effects of driver variability and provided repeatable open-loop data. For most manoeuvres, the driver control inputs were repeated and mean values and 95 per cent confidence intervals were computed. The simulation is driven using mean values.

Test data from the following manoeuvres are compared with simulation predictions: slowly increasing steer, step steer, pulse steer, double lane change, straight line braking and straight line acceleration. Developing simulation models which result in analytical predictions that compare well with experimental measurements is not an easy task, particularly when extreme manoeuvring conditions are considered. For limit performance manoeuvres, only correct trends or relatively close results should be expected, since the mechanics of these extreme conditions (especially tyres) are extremely difficult to model. For NADS, the vehicle simulations are limited to near loss of control conditions, and the simulation is stopped if severe yaw rates (spin-outs) or high roll angles (imminent tip-ups) are predicted.

3.2 Vehicle directional dynamics

3.2.1 Slowly increasing steer

For this manoeuvre, a measured, slowly increasing, handwheel angle (about $10^\circ/\text{s}$) is input to NADSdyna while maintaining constant vehicle speed until maximum

lateral acceleration is achieved. At high steering angles, front tyre force saturation comes into play and lateral acceleration reaches its limit. Simulation predictions matched measured lateral acceleration quite well, confirming appropriate understeer. Simulated roll gradient also matched measured response relatively well. These results indicate appropriate non-linear vehicle model properties as well as appropriate tyre model forces up to saturation.

3.2.2 Step steer input

This manoeuvre is a pseudo-step input of handwheel angle; Fig. 8. It is used to determine steady state as well

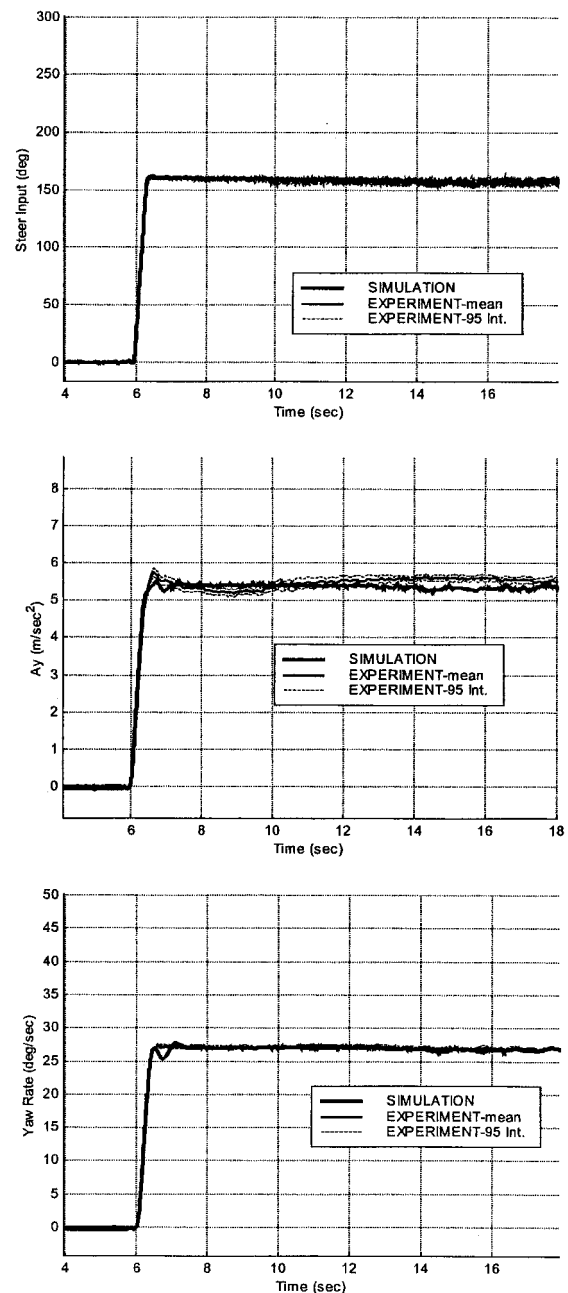


Fig. 8 Simulated and measured step steer—12 m/s

as transient responses. The experimental mean values and 95 per cent confidence limits are shown on the graphs. The timing and peak levels of the vehicle responses for simulation and experiment closely match.

3.2.3 Lane change manoeuvres

Besides evaluation of steady state, transient and frequency response predictions, NADSdyna is checked against manoeuvres that replicate real-world complex scenarios such as lane-change manoeuvres. No statistical analysis was done since only one run was performed for each manoeuvre. The steering input represents driver-vehicle closed-loop performance, with sufficient amplitude to produce moderate to high lateral accelerations. Figure 9 shows that NADSdyna does a good job of predicting timing and peaks of vehicle responses.

3.2.4 Pulse steer frequency response

Pulse steer manoeuvres are used to generate vehicle frequency responses to steering inputs. By selecting a short duration of pulse input, it is possible to excite a full range of frequencies using a single experimental test or simulation run. In field experiments, a pulse with duration of about 0.2 s is generated which produces good steering power up to 3–4 Hz [25].

NADSdyna frequency response predictions were in

generally good agreement with experimental results, particularly below 10 rad/s. Above about 10 rad/s, NADSdyna does not fully capture all of the higher-order dynamics of the actual vehicle. Nonetheless, the predictions are believed to be acceptable for NADS. (The companion paper [19] contains frequency response graphs comparing VDANL simulation with experimental results for the Cherokee.)

3.3 Longitudinal dynamics

3.3.1 Straight-line braking

This manoeuvre is performed from mild to severe conditions. Figure 10 contains results from severe braking runs. The results show that NADSdyna predicted reasonable longitudinal deceleration and longitudinal speed consistent with the deceleration. The brake line pressures were also well predicted by the brake model.

An algorithm was added to the NADSdyna wheel spin dynamics to model kinetic energy and damping considerations at the end of severe braking. This logic provides a smooth transition from a state of high longitudinal tyre forces down to a state of relatively low forces required once the vehicle comes to a complete stop.

3.3.2 Straight-line acceleration

This manoeuvre uses a step throttle input starting from zero vehicle speed. The results show that acceleration, wheel spin rates and vehicle speed are well predicted. The trends in engine speed are similar. Efforts to tune modelled transmission shift points are currently underway. Preliminary testing on NADS has indicated that the current powertrain models provide an acceptable degree of fidelity for use with NADS.

4 CONCLUSIONS

This paper has presented an overview of major components of NADSdyna, the vehicle dynamics simulation used on NADS. The model and parameter measurement processes for a Jeep Cherokee were discussed to represent the methodologies used.

Evaluation of the Jeep Cherokee model (and other vehicles modelled) has shown that NADSdyna predicts fundamental mechanics of vehicle handling responses. Simulation predictions were compared with experimental responses for slowly increasing steer, step steer and lane change manoeuvres. The simulation evaluation included comparisons of vehicle frequency responses for pulse steer tests. In addition, straight-line braking and accelerating manoeuvres were evaluated.

The ultimate evaluation of NADSdyna will come when it is fully used in NADS. Efforts are well underway to incorporate NADSdyna into NADS. NADSdyna

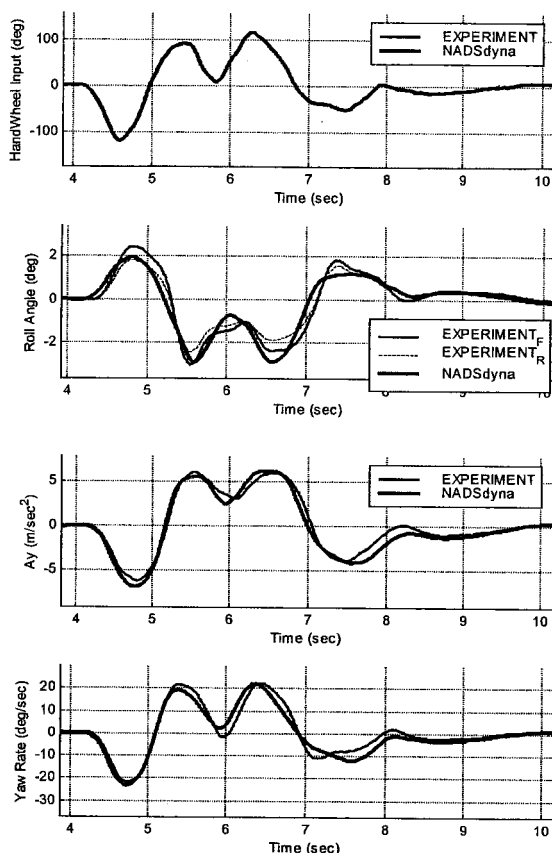


Fig. 9 Simulated and measured lane change—22.5 m/s

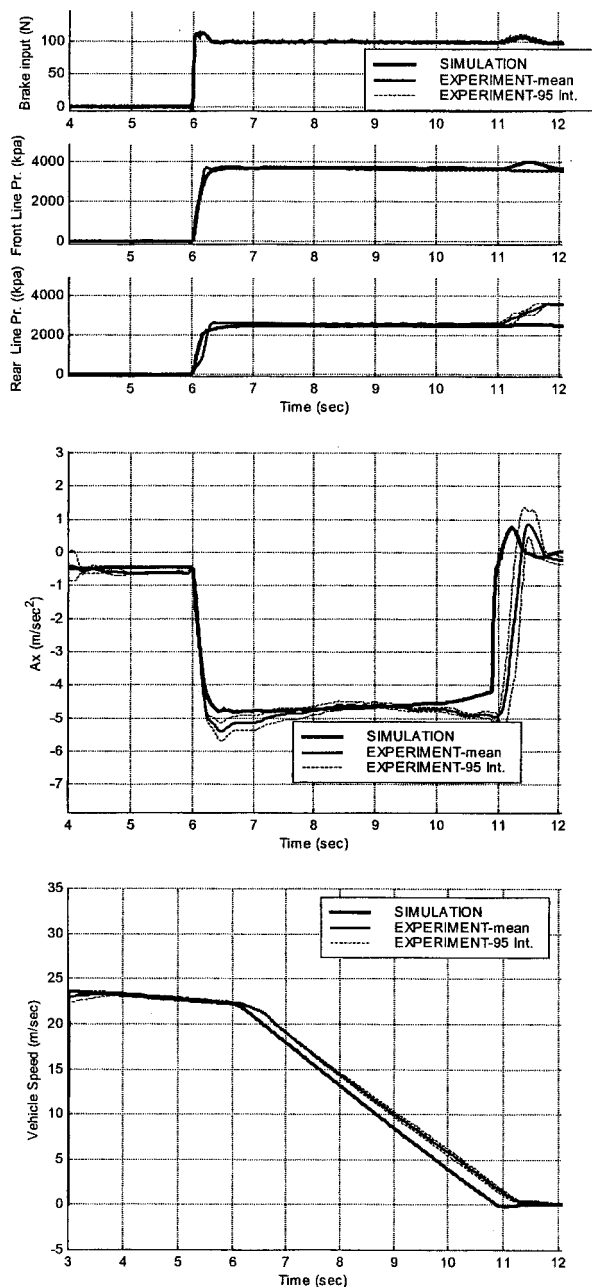


Fig. 10 Simulated and measured severe braking

has been evaluated off line using mostly open-loop manoeuvres and simulation conditions. Tuning of several NADSdyna models and parameters has been necessary as a result of running the simulation in the closed-loop NADS environment. These preliminary studies using NADS suggest that NADSdyna is comprehensive, accurate and fast enough for reliable use in a high fidelity simulator. When all NADS systems, including NADSdyna, are fully integrated and tuned, NADS promises to be the world's most advanced high fidelity driving simulator.

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